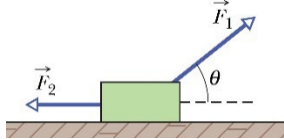


Intro to Forces Practice Problems

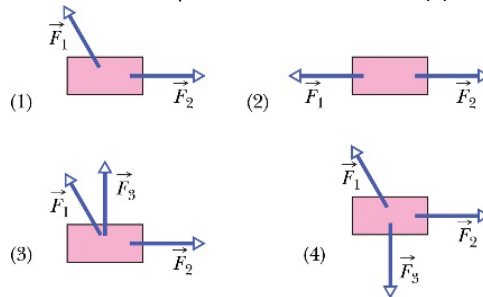
Name: _____

1. In the figure below, forces $\vec{F}_1$ and $\vec{F}_2$ are applied to a box as it slides at a constant velocity over a frictionless floor. We are to decrease the angle θ without changing the magnitude of $\vec{F}_1$. For constant velocity, should we increase, decrease, or maintain the magnitude of $\vec{F}_2$?

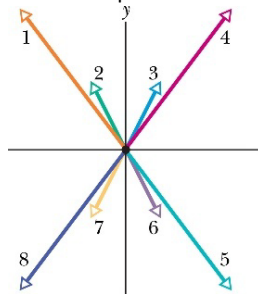


2. At time $t = 0$, constant force $\vec{F}$ begins to act on a rock moving through deep space in the $+x$ direction.
- For time $t > 0$, which are possible functions $x(t)$ for the rock's position:
 - $x = 4t - 3$
 - $x = -4t^2 + 6t - 3$
 - $x = 4t^2 + 6t - 3$
 - For which function is $\vec{F}$ directed opposite the rock's initial direction of motion?

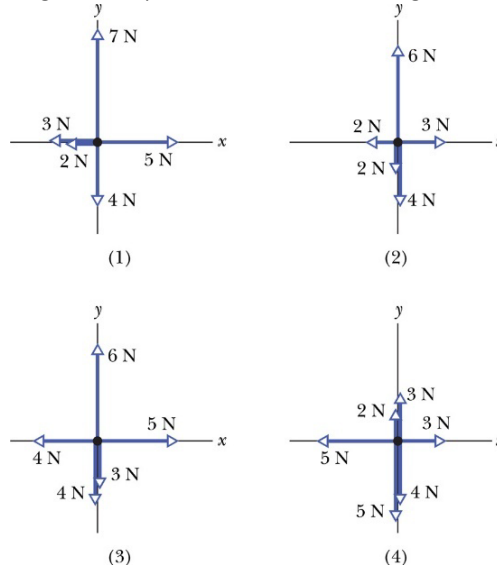
3. The figure below shows overhead views of four situations in which forces act on a block that lies on a frictionless floor. If the force magnitudes are chosen properly, in which situations is it possible that the block is (a) stationary and (b) moving with a constant velocity?



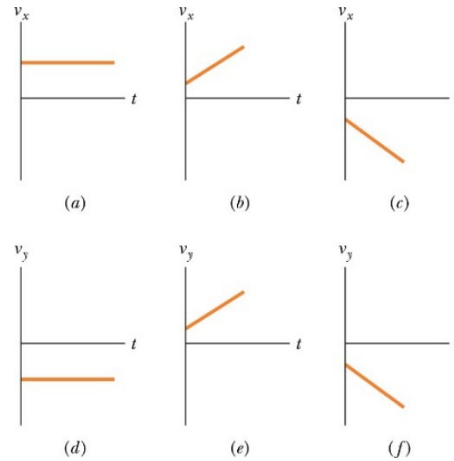
4. Two horizontal forces, $\vec{F}_1 = (3 \text{ N})\hat{i} - (4 \text{ N})\hat{j}$ and $\vec{F}_2 = -(1 \text{ N})\hat{i} - (2 \text{ N})\hat{j}$ pull a banana split across a frictionless lunch counter. Without using a calculator, determine which of the vectors in the free-body diagram below best represent (a) $\vec{F}_1$ and (b) $\vec{F}_2$. What is the net force component along (c) the x axis and (d) the y axis? Into which quadrants do the (e) net-force vector and (f) acceleration vector point?



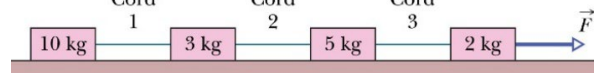
5. The figure below gives the free-body diagrams for four situations in which an object is pulled by several forces across a frictionless floor, as seen from overhead. In which situations does the object's acceleration $\vec{a}$ have (a) an x component and (b) a y component? (c) In each situation, give the direction of $\vec{a}$ by naming either a quadrant or a direction along an axis.



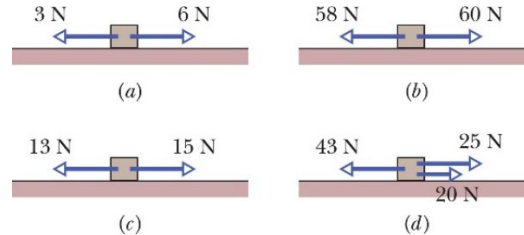
6. The figure below gives three graphs of velocity component $v_x(t)$ and three graphs of velocity component $v_y(t)$. The graphs are not to scale. Which $v_x(t)$ graph and which $v_y(t)$ graph best correspond to each of the four situations in Question 5?



7. The figure below shows a train of four blocks being pulled across a frictionless floor by force $\vec{F}$. What total mass is accelerated to the right by (a) force $\vec{F}$, (b) cord 3, and (c) cord 1? (d) Rank the blocks according to their accelerations from greatest to least. (e) Rank the cords according to their tension from greatest to least.



8. The figure below shows the same object in four situations where horizontal forces are applied. Rank the situations according to the magnitude of the box's acceleration from greatest to least.



9. A vertical force $\vec{F}$ is applied to a block of mass m that lies on the floor. What happens to the magnitude of the normal force $\vec{F}_N$ on the block from the floor as magnitude F is increased from zero if force $\vec{F}$ is (a) downward and (b) upward?

10. The figure below shows four choices for the direction of a force of magnitude F to be applied to a block on an inclined plane. The directions are either horizontal or vertical. (For choices a and b , the force is not enough to lift the block up off of the plane. Rank the choices according to the magnitude of the normal force on the block from the plane from greatest to least.

